Stochastic Write Magnetic Tunnel Junctions for Probabilistic Computing

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Magnetic tunnel junctions (MTJs) are widely employed as nonvolatile memory elements and are emerging as promising hardware building blocks for probabilistic computing. Here we demonstrate on-demand stochasticity in perpendicularly magnetized MTJs that are magnetically stable at room temperature by actuating them with nanosecond electrical pulses in the ballistic spin-transfer torque regime. Unlike superparamagnetic devices that rely on spontaneous thermal fluctuations, these stochastic write MTJs (SW-MTJs) generate tunable stochastic output signals through electrically controlled switching. Our experiments have demonstrated reproducible generation of random bit streams at rates up to 100 MHz per device, as well as tunable random telegraph noise. By integrating individual SW-MTJs with custom readout electronics and FPGA-based control, we realized a true random number generator that passes the complete NIST statistical test suite without post-processing. We also demonstrated that electrically coupled SW-MTJs can realize tunable circuit-mediated interactions that map onto effective Ising couplings, enabling the implementation of interacting Ising networks in hardware. These results establish SW-MTJs as a scalable hardware platform for random number generation, probabilistic computing, and physics-inspired optimization architectures.

***Index Terms*—Magnetic tunnel junctions; spin-transfer torque; true random number generation; probabilistic computing; Ising machines**

I. Introduction

Stochastic write MTJs (SW-MTJs) are based on thermally stable perpendicular magnetic tunnel junctions similar to those used in commercial magnetic random access memory (MRAM) technology. Unlike superparamagnetic MTJs, whose stochastic behavior arises from spontaneous thermal fluctuations, SW-MTJs are driven into a probabilistic switching regime through the application of electrical pulses. A recent review comparing superparamagnetic and stochastic-write approaches can be found in Ref. [1]. In SW-MTJs, the switching probability can be continuously tuned by the pulse amplitude and duration, enabling stochastic behavior on demand while preserving long-term magnetic stability [2]. Nanosecond device operation was first demonstrated in Ref. [3] and subsequently shown to be robust to temperature variations in Ref. [4].

Figure 1 illustrates the operating principle of an SW-MTJ. A nanosecond electrical pulse drives probabilistic switching from a well-defined initial magnetic state, with the switching probability controlled by the pulse amplitude and duration. This enables precise control of the output probability distribution.

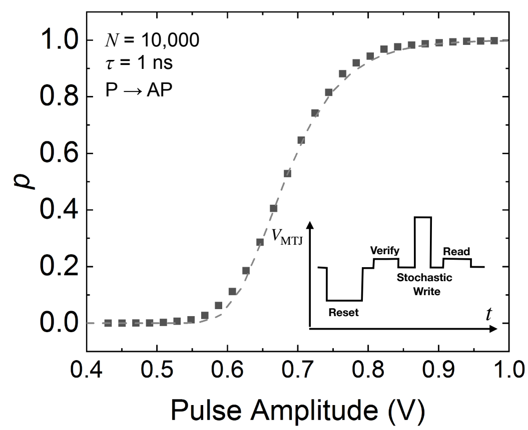

Fig. 1. Switching probability of a stochastic write MTJ (SW-MTJ) as a function of pulse amplitude for a 1 ns write pulse. The MTJ remains thermally stable at room temperature but exhibits electrically tunable switching probabilities. The inset illustrates the pulse sequence used for stochastic operation. Adapted from Ref. [3].

II. Random Number Generation and Tunable Stochastic Signals

The controlled stochasticity of SW-MTJs enables their use as compact entropy sources for true random number generation. Initial demonstrations showed that bitstreams generated at repetition rates of 1 kHz and conditioned using an XOR operation satisfied NIST randomness criteria [3]. Subsequent work demonstrated high-speed generation of random numbers with bit rates greater than 100 MHz per device [5].

Recent FPGA-based implementations integrated SW-MTJs with custom electronics to realize real-time random bit generation. In one implementation, more than one trillion random bits were generated and validated using standard statistical randomness tests; after XOR post-processing, the resulting bitstreams passed the NIST statistical test suite [6]. More recently, FPGA-based feedback and control circuitry was used to generate bitstreams that could be streamed directly to a host computer and pass the complete NIST statistical test suite without any post-processing [7]. These results demonstrate the suitability of SW-MTJs as practical hardware entropy sources for computing applications.

In addition to binary random bit generation, electrically actuated MTJs can be operated in a regime that produces tunable random telegraph noise. In this mode, the pulse sequence is modified to consist of stochastic write pulses of alternating polarity [8], enabling probabilistic transitions between the parallel and antiparallel magnetic states (Fig. 2). The

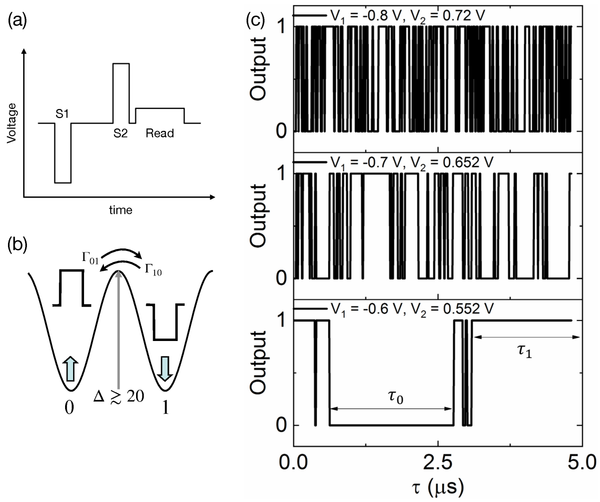

Fig. 2. Generation of tunable random telegraph noise using a SW-MTJ. (a) Pulse sequence consisting of stochastic write pulses S1 and S2 of alternating polarity followed by state readout. (b) Schematic energy landscape illustrating probabilistic transitions between the parallel (0) and antiparallel (1) states with transition probabilities Γ_{01} and Γ_{10}. (c) Representative telegraph signals obtained for different pulse amplitudes. The average dwell times and state occupancies can be electrically tuned over many orders of magnitude, reaching the nanosecond regime through adjustment of the pulse sequence, enabling controllable stochastic dynamics for probabilistic and neuromorphic computing applications. Figure adapted from Ref. [9].

fluctuation rates and state occupancies can be adjusted through the pulse amplitudes, durations, and repetition rates, providing a controllable stochastic signal source that is attractive for unconventional and neuromorphic computing architectures [9].

III. Coupled Stochastic MTJs for Probabilistic Computing

Beyond random number generation, SW-MTJs can serve as interacting stochastic elements. Electrical coupling between devices produces circuit-mediated interactions that map naturally onto effective Ising couplings (Fig. 3). Recent experiments demonstrated electrically connected stochastic MTJs whose collective behavior can be controlled through circuit design and operating conditions [10]. These results establish a pathway toward hardware implementations of Ising networks, probabilistic graphical models, and physics-inspired optimization architectures.

IV. Summary

SW-MTJs combine room-temperature magnetic stability with electrically actuated stochasticity, enabling high-speed random number generation, tunable stochastic signal generation, and programmable interactions between devices. These capabilities establish SW-MTJs as a promising hardware platform for probabilistic computing and physics-inspired optimization systems.

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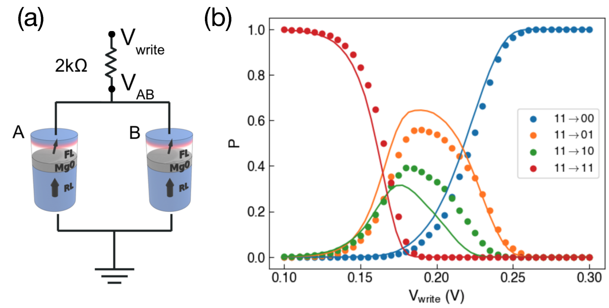

Fig. 3. Electrical coupling between SW-MTJs. (a) Experimental configuration consisting of two SW-MTJs connected in parallel to a series resistor. (b) Probabilities of the four possible two-bit states as a function of write pulse amplitude following initialization in the 11 state. The coupled switching dynamics produce tunable antiferromagnetic correlations between the MTJs and demonstrate a circuit-mediated interaction that can be mapped onto an effective Ising coupling. Figure adapted from Ref. [10].

Sidi El Valli, Jonathan Z. Sun, Michael Tsao and other collaborators whose contributions are reflected in the cited publications. This work was supported in part by the Office of Naval Research under Grant No. N00014-23-1-2771.

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